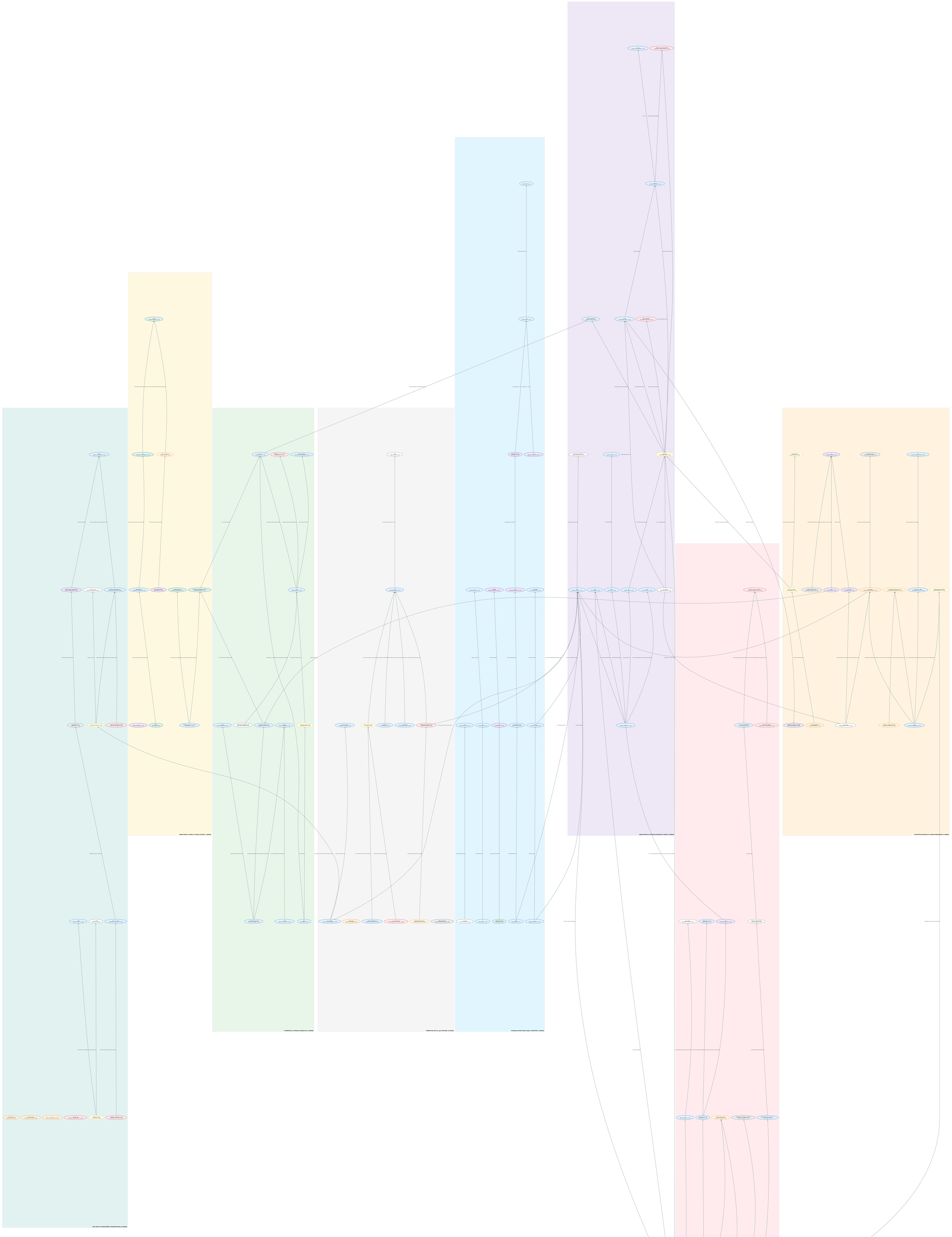

Module 5: Symmetries, Anomalies & Gauge Dynamics
N5.5.4: Higgs Mechanism Local gauge SSB → eaten bosons
N5.3.1: Gauge Fixing Condition G[A] = 0
N5.4.3: Slavnov-Taylor Identities Non-Abelian Ward identities
N5.5.1: Symmetry-Breaking Potential Degenerate Vacua
N5.2.7: Gauge Redundancy Orbit overcounting
N5.3.2: Faddeev-Popov Det Δ_FP
N5.4.1: Ward-Takahashi Identity Quantum conservation
N5.5.2: Goldstone's Theorem Continuous global SSB
N5.1.3: Noether Current J^μ(x)
N5.3.3: Ghost Fields c, c̄ (Unphysical anticommuting)
N5.4.4: Quantum Anomaly Symmetry breaking on quantization
N5.1.4: Noether's Theorem Classical conservation
N5.2.2: Gauge Field A_μ^a (Connection)
N5.1.5: Conserved Charge Q
N5.3.4: BRST Symmetry Residual global symmetry
N5.2.3: Covariant Derivative D_μ
N5.2.4: Field Strength F_μν^a (Curvature)
N5.1.6: Charge as Generator [Q, Φ] = δΦ
N5.4.6: Gauge Anomaly Cancellation Consistency criterion
N5.3.5: BRST Charge Q_B (Nilpotent Q_B^2 = 0)
N5.3.6: Physical BRST Cohomology ℋ_phys = ker Q_B / im Q_B
Module 3: Quantization Routes & Functional Generators
N3.2.2: Generating Functional Z[J]
N3.3.3: LSZ Reduction Formula Green's functions → S-Matrix
N3.1.2: Commutation Relations [Φ, Π] = iħδ
N3.1.7: Heisenberg Picture Operators evolve
N3.1.1: Canonical Quant Map Classical → Quantum operators
N3.2.6: Schwinger-Dyson Eq Quantum EOM functional
N3.2.4: Connected Generator W[J] = -iħ ln Z[J]
N3.3.1: Operator/PI Correspondence VEV ↔ Functional derivative
N3.2.1: Path-Integral prescription ∫𝒟Φ e^(iS)
N3.1.3: Mode Operators a_p, a_p^†
N3.1.6: Time Ordering T[𝒪]
N3.1.12: Ordering Ambiguities High-energy UV signal
N3.1.10: Heisenberg Eq d𝒪/dt = i[ℋ, 𝒪]
N3.2.5: Quantum Effective Action Γ[φ]
N3.2.8: Path Integral Measure 𝒟Φ (formal limit)
N3.1.5: Normal Ordering :𝒪:
N3.1.11: Quantum EOM Operator EOM
Module 2: Action, Variational Structure & Classical EOM
N2.1.2: Lagrangian Density ℒ
N2.3.6: Poisson Brackets {Φ, Π} = δ
N2.1.5: Interaction Term ℒ_int
N2.2.1: Action Variation δS
N2.1.4: Kinetic Term ℒ_kin
N2.1.6: Potential Term V(Φ)
N2.1.1: Action Functional S[Φ]
N2.2.3: Euler-Lagrange Eq Field EOM
N2.1.8: Mass Parameter m
N2.1.9: Dimensional Analysis Engineering dimensions
N2.2.4: Boundary Surface Term Asymptotic falloff
N2.3.1: Classical Momentum Π_cl
N2.2.6: Green's Functions Differential inversion
N2.3.2: Hamiltonian Density ℋ = Πφ̇ - ℒ
N2.3.5: Singular Hessian Constraints Gauge system (Dirac analysis)
N2.3.3: Hamiltonian Functional H = ∫ℋ
Module 1: Spacetime, Field & State Space Primitives
N1.1.1: Spacetime Manifold M
N1.2.2: Classical Field Φ_cl(x)
N1.4.8: Observable Hermitian operator
N1.3.1: Hilbert Space ℋ
N1.1.12: Poincaré Group ISO(3,1)
N1.1.6: Metric Structure g_μν(x)
N1.2.10: Quantum Field Φ(x) (Distribution)
N1.4.5: Local Operator 𝒪(x)
N1.3.2: State Vector |Ψ⟩
N1.1.13: Minkowski Space ℝ^d
N1.1.9: Causal Structure Light Cone
N1.4.2: Canonical Momentum Π(x)
N1.4.6: Composite Operator 𝒪[Φ](x)
N1.3.3: Density Operator ρ
N1.4.4: Momentum Operator P_μ
N1.4.3: Hamiltonian H (Time translation)
N1.3.5: Vacuum State |0⟩
N1.3.6: Fock Space ℱ
Module 9: Wilsonian EFT & Scale Decoupling
N9.6.1: Non-Perturbative Matching Unresolved bridge
N9.3.2: Decoupling Theorem Heavy mass decoupling
N9.5.3: Scheme-scale independence d O_obs / dμ = 0
N9.4.2: Anomalous Dim Matrix γ_ij (Operator mixing)
N9.1.2: Momentum Cutoff Λ (EFT scale)
N9.1.1: Wilsonian Effective Action S_eff[Φ_<; Λ]
N9.3.1: Matching Condition E_low amplitudes equivalence
N9.5.2: Low-Energy Response ∂O_obs / ∂c_i
N9.5.1: Response of c_i ∂c_i / ∂g_UV
N9.4.1: RGE for c_i μ d c_i / dμ = γ_ij c_j
N9.1.5: Effective Lagrangian ℒ_eff = Σ c_i O_i
N9.2.1: Wilson Coefficients c_i(Λ)
N9.2.2: Operator Basis {O_i}
Module 4: Correlation Functions & Propagators
N4.3.4: Self-Energy Σ (1PI)
N4.4.2: Commutator Function Δ(x-y)
N4.1.1: Wightman Function Non-time-ordered VEV
N4.3.5: Dyson Equation G_F = G_0 + G_0 Σ G_F
N4.1.3: Retarded Function G_R(x,y)
N4.1.2: Feynman Propagator Time-ordered two-point
N4.4.4: Källén-Lehmann Rep Full propagator spectral form
N4.1.4: Advanced Function G_A(x,y)
N4.2.5: Spectral Density ρ(s)
N4.2.6: Field Renormalization Residue Z (Residue of pole)
N4.4.1: Microcausality [O_1, O_2] = 0 (Spacelike)
N4.2.1: Free Propagator G_F^(0)
Module 7: Spectral Theory, S-Matrix & Observables
N7.6.1: Complex Pole m_R^2 - i m_R Γ_R (Unstable)
N7.4.5: Decay Rate Γ_decay
N7.2.4: Scattering S-Matrix S
N7.1.5: Physical Pole Mass m_phys (Gauge/Scheme invariant)
N7.6.2: Breit-Wigner shape Resonance shape
N7.5.4: Cutkosky Rules Imaginary loop cuts
N7.4.1: Scattering Amplitude ℳ (Amputated)
N7.5.2: Optical Theorem Im ℳ(s) = s σ_tot
N7.4.3: Differential Cross-Section dσ
N7.4.4: Total Cross-Section σ_tot
Module 6: Renormalization, Regularization & Scale Flow
N6.4.5: Power-Counting Renorm D(Γ) ≤ d (Relevant / Marginal)
N6.3.2: Beta Function β(g) = μ dg/dμ
N6.2.7: Renormalization Conditions OS, MS, MS̄
N6.2.5: Renormalization Scale μ
N6.2.4: Renormalized Params Φ_R, m_R, g_R
N6.2.1: Bare Quantities Φ_0, m_0, g_0
N6.2.6: Counterterms Lagrangian ℒ_CT = ℒ_0 - ℒ_R
N6.3.6: Fixed Points β(g*) = 0
N6.3.4: Running Coupling g(μ)
N6.5.1: S-Matrix Scheme Indep Physical predictions invariant
N6.3.1: Callan-Symanzik Eq
N6.1.1: UV Divergence Loop singularities
N6.2.8: Renormalized Correlators G_R (Regulator removed)
N6.3.8: Scale Trace Anomaly ∂_μ T^μ_μ ≠ 0
N6.1.3: Regularization Scheme Dimensional / Cutoff / Lattice
N6.1.8: Regularized Correlator G_reg